FEDERAL SPECIFICATION

BEARING, SLEEVE, (BRONZE, PLAIN OR FLANGED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tubular-shaped, plain or flanged, bronze bearings.

1.1.1 *Federal specification coverage.*—Federal specifications do not include all types of commodities indicated by the titles of the specifications, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.*—The bearings shall be of the following types as specified (see 6.2):

Type I.—Bearing, plain (see figure 1).

Type II.—Bearing, single flanged (see figure 2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN-P-515—Plywood, Container Grade.
QQ-L-225—Leaded Tin Bronze Castings and High Leaded Tin Bronze Castings.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.
PPP-B-640 — Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

Military Standards:

MIL-STD-10 — Surface Roughness, Waviness and Lay.

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MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification, Part 4.

(Application for copies should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

National Motor Freight Classification No. A–3 and No. 13.

(Application for copies should be addressed to American Trucking Associations, Inc., Traffic Department, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

Parcel Post Regulations.

(Application for copies should be obtained from any first class U. S. Post Office.)

3. REQUIREMENTS

3.1 Material.—The bronze material used in the manufacture of the bearings shall be in conformance with composition 12 of Federal Specification QQ–L–225.

3.2 Design.—The bearings shall be designed to reduce friction and carry a kinetic load on the surface parallel to the axis of the bore.

3.3 Dimensions.—Dimensions shall be within the tolerances specified in figures 1 or 2 as applicable. Where no tolerance is given, angles shall be within ± 2° and dimensions shall be within ± 0.010-inch.

3.4 Finish.—The inside and outside diameters of both bearings and the flange surfaces shall not exceed the average finish of 32 r.m.s. (root mean square) surface roughness as specified in Military Standard MIL–STD–10.

3.5 Concentricity. — Type I and type II bearing diameters shall be concentric within 0.002-inch in total indicator reading for all sizes.

3.6 Workmanship.—The bearings shall be of uniform quality and structure and shall be free from all defects (dirt, dross, oxides, porous or spongy spots, shrinkage cavities, cracks, cold-shuts, blowholes, hard spots, blisters or other injurious defects.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibilities. — Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Inspection.—Unless otherwise specified by the contracting officer, or his representative, inspection shall be in accordance with Military Standard MIL–STD–105.

4.2.1 *Lot formation.*—For examination in accordance with 4.2.2 a lot shall consist of all bushing of one size and type produced at one place of manufacture for one contract or order and presented for inspection at the same time. A lot for inspection of preparation for delivery shall be as specified in Military Specification MIL–P–116.

4.2.2 *Examination.*—The classification of defects in tables I, II, and III shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.2.1 *Resubmission.* — All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the

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corrected action taken, the lot may be resubmitted for acceptance inspection as outlined in Military Standard MIL–STD–105.

4.2.3 *Inspection equipment.*—Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contracted requirements. The Government reserves the right to certify all such equipment and to use this equipment for Government verification inspection.

4.2.4 *Material.*—Certified reports showing chemical compositions and physical properties shall be furnished by the contractor for each lot of castings to determine compliance with 3.1.

4.2.5 *Dimensions.*—Dimensional inspection of each sample bearing shall be conducted to verify compliance with this specification regarding inside and outside diameters, flange thickness, concentricity and such other dimensions as may be specified in the contract or order (see 3.3 and 6.2(b)).

4.2.6 *Finish.*—The surface roughness shall be inspected in accordance with the procedures outlined in Military Standard MIL–STD–10 (see 3.4.)

4.2.7 *Concentricity.*—The sample bearings shall be inspected for concentricity of its diameters by use of a fixture or set-up with a dial indicator attached to determine compliance with 3.5.

4.2.8 *Workmanship.* — Visual examination shall be made to determine compliance with 3.6.

4.3 Test methods and procedures.

4.3.1 *Packaging tests.* — Each package in the applicable sample shall be subjected to the applicable packaging tests of Military Specification MIL–P–116.

4.3.2 *Determination of cleanliness test.*— Periodically the applicable surfaces of one or more units shall be subjected to the "Determination of Cleanliness" tests in accordance with Military Specification MIL–P–116.

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.—Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

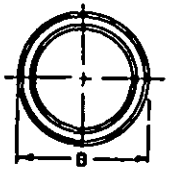
5.1.1.1 *Cleaning.*—Each bearing shall be cleaned to remove dirt, smudges, fingerprints and other foreign substances by process C–5 of Military Specification MIL–P–116.

5.1.1.2 *Drying.* — The bearings shall be dried in accordance with any drying procedure specified in Military Specification MIL–P–116.

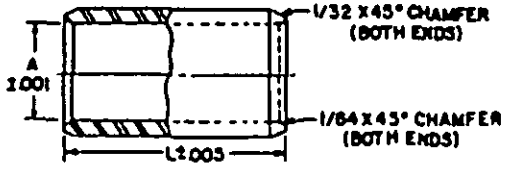
TABLE I.—*Classification of defects*
(Use inspection level I, and sampling plan table IV–A of Military Standard MIL–STD–105)

Major	AQL — 2.5 percent	Requirement paragraph	Inspection paragraph	Inspection method
1.	Dimensions	3.3	4.2.5	SME (see 6.5)
2.	Finish	3.4	4.2.6	SME
3.	Concentricity	3.5	4.2.7	SME
4.	Workmanship	3.6	4.2.8	Visual

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TYPE I

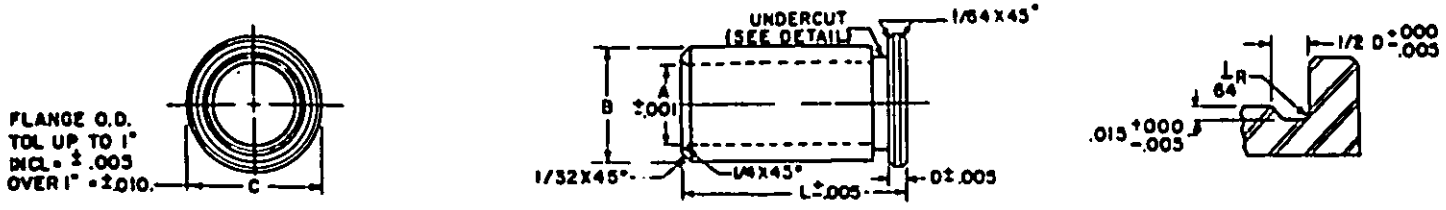


BEARING, SLEEVE, BRONZE, PLAIN

A. BODY, INSIDE DIA.	3/16	1/4	5/16	3/8	7/16	1/2	9/16	5/8	11/16	3/4	7/8	1	1 1/8	1 1/4	1 3/8	1 1/2	1 3/4	2	2 1/4	2 1/2	2 3/4	3	3 1/2	4
B. BODY, OUTSIDE DIA. (MAX.	.378	.4405	.503	.5655	.628	.6905	.753	.8155	.878	1.003	1.128	1.253	1.378	1.503	1.6905	1.8155	2.0655	2.378	2.628	2.878	3.130	3.380	4.005	4.505
(MIN.	.377	.4395	.502	.5645	.627	.6895	.752	.8145	.877	1.002	1.127	1.252	1.377	1.502	1.6895	1.8145	2.0645	2.377	2.502	2.877	3.128	3.378	4.003	4.503
WALL THICKNESS (NOM	3/32	3/32	3/32	3/32	3/32	3/32	3/32	3/32	3/32	1/8	1/8	1/8	1/8	1/8	5/32	5/32	5/32	3/16	3/16	3/16	3/16	3/16	1/4	1/4
L. LENGTH	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.	SIZE NO.
1/2	1	8	15	25	34	46	55	65	75															
5/8	2	9	16	26	35	47	56	66	76	85	96													
3/4	3	10	17	27	36	48	57	67	77	86	97	106	116											
1		11	18	28	37	49	58	68	78	87	98	107	117	127	138									
1 1/4		12	19	29	38	50	59	69	79	88	99	108	118	128	139	150	171							
1 1/2								70	80	89	100	109	119	129	140	151	172	182						
1 3/4								71	81	90	101	110	120	130	141	152	173	183	193					
2								72	82	91		111	121	131	142	153	174	184	194	203	213	223		
2 1/4													122	132	143	154	175	185	195	204	214	224		
2 1/2														133	144	155	176	186	196	205	215	225	234	
2 3/4																156	177	187	197	206	216	226	235	
3																		188	198	207	217	227	236	245
3 1/2																				208	218	228	237	246
4																						229	238	247
4 1/2																							239	248
5																								249

FIGURE 1

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TYPE 2

BEARING, SLEEVE, BRONZE, FLANGED

A. BODY, INSIDE DIA.	1/4	5/16	3/8	7/16	1/2	9/16	5/8	11/16	3/4	7/8	1	1 1/8	1 1/4	1 3/8	1 1/2	1 3/4	2	2 1/4	2 1/2	2 3/4	3	3 1/2	4
B. BODY, OUTSIDE DIA. (MAX)	.4405	.503	.5655	.628	.6905	.753	.8155	.878	1.003	1.128	1.253	1.378	1.503	1.6905	1.8155	2.0655	2.378	2.628	2.878	3.130	3.380	4.005	4.505
B. BODY, OUTSIDE DIA. (MIN)	.4395	.502	.5645	.627	.6895	.752	.8145	.877	1.002	1.127	1.252	1.377	1.502	1.6893	1.8143	2.0643	2.377	2.627	2.877	3.126	3.378	4.003	4.503
C. FLANGE DIA.	9/16	5/8	11/16	3/4	13/16	15/16	1	1 1/16	1 1/4	1 3/8	1 1/2	1 5/8	1 3/4	2	2 1/8	2 3/8	2 3/4	3	3 1/4	3 1/2	3 3/4	4 1/2	5
D. FLANGE THICKNESS	3/32	3/32	3/32	3/32	3/32	3/32	3/32	3/32	1/8	1/8	1/8	1/8	1/8	5/32	5/32	5/32	3/16	3/16	3/16	3/16	3/16	1/4	1/4
L. LENGTH	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO	SIZE NO
5/16	1	8																					
3/8	2	9	17	26	36																		
7/16	3	10	18	27	37																		
1/2		11	19	28	38																		
9/16		12	20	29	39																		
5/8			21	30	40	47	56	65															
3/4				31	41	48	57	66	75	84													
7/8					42	49	58	67	76	85	94	104											
1						50	59	68	77	86	95	105	114										
1 1/8							60	69	78	87	96	106	115	124									
1 3/8								70	79	88	97	107	116	125	133	142							
1 5/8										89	98	108	117	126	134	143	152						
1 7/8											99	109	118	127	135	144	153	162					
2 1/8													119	128	136	145	154	163	171	181	191		
2 3/8															137	146	155	164	172	182	192		
2 5/8																147	156	165	173	183	193		
3 1/8																	157	166	174	184	194		
3 1/4																			175	185	195	202	209
3 5/8																			176	186	196	203	210
4 1/4																					197	204	211
5 1/4																							212

FIGURE 2

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TABLE II.—*Classification of defects*
(Use inspection level I, and sampling plan table IV–A of Military Standard MIL–STD–105)

Major AQL — 2.5 percent	Requirement paragraph	Test method
1. Markings illegible, missing (levels A, B and C)	5.3	Visual
2. Greaseproof, waterproof bag, damaged or improperly sealed (level A)	5.1.1.4	Visual
3. Improper cleaning (level A)	5.1.1.1	Visual
4. Improper drying (level A)	5.1.1.2	Visual
5. Item not adequately coated with correct type of preservative (level A)	5.1.1.3	Visual
Minor AQL — 4.0 percent		
41. Unit quantities not as specified (level A)	5.1.1.4	Visual
42. Unit packaging (level A)	5.1.1.4	Visual
43. Defective taping or sealing of fiberboard container (level A)	5.1.1.5	Visual
44. Packing (levels A, B and C)	5.2	Visual
45. Improper strapping (level A)	5.2.1	Visual

TABLE III.—*Classification of defects*
(Use inspection level L7, and sampling plan table IV–A of Military Standard MIL–STD–105)

Major AQL — 2.5 percent	Requirement paragraph	Inspection paragraph	Test method
Heat Seal (Level A)	5.1.1.4	4.3.1	Visual
Quick Seal (Level A)	5.1.1.4	4.3.1	Visual

5.1.1.3 *Preservative application.* — Each bearing shall be coated with type P–2 preservative as specified in Military Specification MIL–P–116.

5.1.1.4 *Unit packaging.* — The bearings shall be packaged in accordance with method IC–1 of Military Specification MIL–P–116. Identical bearings (bearings of the same stock number) shall be packaged in quantities as specified in table IV.

TABLE IV.

Outside diameter (in inches)	Length (in inches)	Unit quantity
1/8	1/4 to 1 1/4	5
1/2	1/4 to 1 1/4	5
9/16	1/4 to 1 1/4	5
5/8	1/4 to 1 1/4	5
11/16	1/4 to 1 1/4	5
3/4	1/4 to 1 3/4	5
13/16	1/2 to 1 3/4	5
7/8	1/2 to 1 3/4	5
1	1/2 to 1 3/4	5
1 1/8	1/2 to 1 3/4	5
1 1/4	3/4 to 2 3/4	5
1 3/8	3/4 to 2 3/4	5
1 1/2	3/4 to 2 1/4	5
1 1/2	2 1/2 to 2 3/4	1
1 11/16	3/4 to 2	5
1 11/16	2 1/4 to 2 3/4	1
1 13/16	3/4 to 1 3/4	5
1 13/16	2 to 2 3/4	1
2 1/16	1 1/4 to 1 1/2	5
2 1/16	1 1/4 to 3 5/8	1
2 3/16	1 1/4 to 3 5/8	1
2 3/16	1 1/4 to 3 5/8	1
2 1/8	1 1/4 to 3 5/8	1
3 1/8	1 1/4 to 3 5/8	1
3 1/8	1 1/4 to 3 5/8	1
3 5/8	2 to 5 1/4	1
4	2 to 5 1/4	1
4 1/2	2 to 5 1/4	1
4 3/4	2 1/8 to 5 1/4	1

5.1.1.5 *Intermediate packaging.*—Identical bearings (bearings of the same stock number) shall be further packaged in a close-fitting intermediate container conforming to Federal Specification PPP–B–636, class 2, W5c. The box shall be closed with strips of 2-inch tape conforming to Federal Specification PPP–T–76. The gross weight shall not exceed approximately 20 pounds.

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5.1.2 *Level C.*—Cleaning, drying and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.—Packing shall be level A, B, C, or D as specified (see 6.2).

5.2.1 *Level A.*—Identical bearings (bearings of the same stock number) shall be packed in a close-fitting wirebound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nailed wood box or a fiberboard box conforming to Federal Specification PPP–B–585, class 3, PPP–B–591 (overseas type), PPP–B–601, style A or B, using plywood conforming to Federal Specifications NN–P–515, type II or III, class 2, PPP–B–621, class 2 and PPP–B–636, class 2 or 3, respectively. Closure and strapping requirements shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 150 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 *Level B.*—Identical bearings (bearings of the same stock number) shall be packed in a close-fitting wirebound wood box, wood-cleated fiberboard box, wood-cleated-plywood box, nailed wood box or fiberboard boxes conforming to Federal Specification PPP–B–585, class 1, PPP–B–591 (domestic type), PPP–B–601 (domestic type), PPP–B–621, class 1, PPP–B–636, class 1 and PPP–B–640, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 150 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* — The bearings shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification, Motor Freight Regulations or Parcel Post Regulations, whichever is applicable.

5.2.4 *Level D.* — The bearings shall be packed for ocean shipment and covered storage at destination where the end use is to be at the initial receiving activity. This level in general conforms to applicable carrier rules and regulations and may be supplier's commercial overseas practice when such meets the requirements.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each unit, intermediate package, and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — Unit packages, intermediate packages, and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. — The bearings are intended for use where moderate friction and bearing play at low rotational speeds are not objectionable, where wear is not excessive and where bearings are not subjected to excessive vibratory shocks or alternating loads.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type, size of diameters and length of bearing (see 1.2).
(*c*) Selection of applicable levels of packaging and packing (see section 5).
(*d*) Marking, if different (see 5.3).

6.3 Military standards. — Items procured under this specification for military use are to be limited to the variety shown on the applicable Military Standards. Personnel of the

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military departments are requested to refer to these documents for guidance.

6.4 Transportation description.—Transportation description applicable to this item is:

Carload minimum weight 30,000 pounds.

Motor volume minimum weight 30,000 pounds.

6.5 Definitions.

6.5.1 *Standard measuring equipment (SME).* — Standard measuring equipment (SME) includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Ordnance
Air Force.

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